THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · EASY LEVEL CHAPTER TEST SET 1

Chapter 8: **Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Easy level: direct, single-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. For the sequence $t_n = 2n + 1$, the value of t_5 is: 1

(a) 13 **(b)** 11 **(c)** 10 **(d)** 12

2. The common difference of the AP $4, 7, 10, 13, \ldots$ is: 1

(a) 2 **(b)** 3 **(c)** 5 **(d)** 4

3. The next term of the AP $2, 7, 12, 17, \ldots$ is: 1

(a) 22 **(b)** 21 **(c)** 23 **(d)** 24

4. The next term of the GP $1, 2, 4, 8, \ldots$ is: 1

(a) 17 **(b)** 15 **(c)** 16 **(d)** 18

5. The common ratio of the GP $3, 6, 12, 24, \ldots$ is: 1

(a) 3 **(b)** 1 **(c)** 2 **(d)** 4

6. The sum of the first 6 natural numbers is: 1

(a) 20 **(b)** 21 **(c)** 22 **(d)** 27

7. The sequence $2, 4, 6, 8, \ldots$ is: 1

(a) a GP **(b)** an AP

(c) neither an AP nor a GP **(d)** both an AP and a GP

8. The 6th term of the sequence $1, 4, 9, 16, \ldots$ is: 1

(a) 36 **(b)** 35 **(c)** 37 **(d)** 42

9. For the sequence $t_n = 3n + 2$, the value of t_4 is: 1

(a) 14 **(b)** 15 **(c)** 13 **(d)** 16

10. The common difference of the AP $10, 15, 20, 25, \ldots$ is: 1

(a) 5 **(b)** 7 **(c)** 6 **(d)** 4

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Write the first four terms of the sequence $t_n = 3n + 1$. 2

12. Find the common difference of the AP $6, 10, 14, 18, \ldots$ and write the next two terms. 2

13. Find the common ratio of the GP $2, 10, 50, \ldots$ and its fourth term. 2

14. Write the first four terms of the sequence $t_n = 2n - 1$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the 10th term of the AP with first term 3 and common difference 4. **3**

16. Find the sum of the first 20 natural numbers. **3**

17. Find the 6th term of the GP with first term 1 and common ratio 2. **3**

OR

Find the 10th term of the AP with first term 5 and common difference 2. **3**

18. Find the sum of the first 30 natural numbers. **3**

OR

Find the 6th term of the GP with first term 2 and common ratio 2. **3**

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A sequence is given by $t_1 = 4$ and $t_{n+1} = t_n + 3$. Write the first five terms, say what kind of sequence it is and find t_{10}. **5**

OR

An AP has first term 2 and common difference 3. Find the sum of its first 10 terms. **5**

20. A sequence is given by $t_1 = 7$ and $t_{n+1} = t_n + 5$. Write the first five terms, say what kind of sequence it is and find t_{10}. **5**

OR

An AP has first term 5 and common difference 4. Find the sum of its first 8 terms. **5**

END OF QUESTION PAPER — Total: 40 Marks

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